

RAVI KUMAR A

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PROFESSIONAL SUMMARY

High-achieving M.Tech Scholar in **Artificial Intelligence & Data Science (8.4 CGPA)** and recipient of the **IEEE Best Paper Award (2025)**. Expert in **Deep Learning (YOLOv8, CNN-LSTM)** and **Full-stack Development**, with a proven record of accelerating model inference to **32 FPS** and enhancing platform engagement benchmarks by **25%**. Technical specialist in architecting scalable AI solutions and cross-platform management systems using Python, React.js, and SQL Server.

TECHNICAL EXPERTISE

- **Languages:** Python (PyTorch, TensorFlow, Keras), Java (Swing/AWT), JavaScript (ES6+), SQL, C.
- **AI & Machine Learning:** Object Detection (YOLOv8/v5), NLP (LSTM, InceptionV3), Reinforcement Learning, Computer Vision, Data Augmentation, Hyperparameter Tuning.
- **Development:** React.js, PHP, Flask, FastAPI, HTML5, CSS3, Bootstrap, Java Desktop Applications.
- **Tools & Databases:** Git, MySQL, SQL Server, Google Colab, Anaconda, XAMPP, Postman.

PROFESSIONAL EXPERIENCE

Software Developer Intern | Silo Fortune Pvt Ltd

Oct 2025 – Present

- Spearheaded a machine learning integration for the **Kissan Sampurna** agricultural platform, deploying recommendation engines that expanded user retention by **25%**.
- Engineered and validated disease classification models for **15+ plant species**, achieving a **94.2% validation accuracy** through strategic data balancing.
- Automated the preprocessing of a **25,000+ image dataset**, reducing end-to-end pipeline latency by **18%**.

Cyber Security Intern | Honeywell

Aug 2023 – Oct 2023

- Performed exhaustive security audits on **12 enterprise modules**, identifying **5 high-risk vulnerabilities** including SQL injection and XSS flaws.
- Facilitated the deployment of critical security patches, effectively hardening the legacy infrastructure and reducing the attack surface by **15%**.

KEY PROJECTS

Underwater Plastic Trash Identification (B.E. Main Project)

Jan 2024 – May 2024

- Developed a real-time marine debris detection system using **YOLOv8**, achieving a sustained **32 FPS** and a **mAP@50 of 63%** on the Trash-ICRA19 dataset.
- Improved detection recall in low-visibility environments from 45% to **60%** via custom noise-reduction and data augmentation scripts.
- Architected a Flask-based monitoring dashboard, streamlining manual video analysis for researchers by **40%**.

Image Caption Generator (Mini Project)

Aug 2023 – Dec 2023

- Designed a multimodal CNN-LSTM architecture (InceptionV3) to process **8,000+ images**, attaining a **BLEU-1 score of 60.44**.
- Optimized the inference pipeline for real-time performance using a **Gradio** interface, achieving a response time of **0.8s**.

Dairy Farm Shop Management System (DBMS Project)

Jan 2023 – Jun 2023

- Built a robust desktop application using **Java and SQL Server** to automate inventory, product tracking, and sales analytics.
- Implemented "Between Dates" reporting logic, eliminating **20%** of manual accounting overhead for shop administrators.

- Optimized database schema for 100+ categories, ensuring sub-second query performance for high-volume transactions.

Online Service Management System (Mini Project)

Aug 2022 – Dec 2022

- Developed a full-stack household service marketplace using **React.js and PHP**, featuring **6 core operational modules**.
- Accelerated provider discovery and search efficiency by **30%** through strategic SQL indexing and normalization.

EDUCATION

M.Tech. AI & Data Science	Malnad College of Engineering	8.4 CGPA	<i>Exp. May 2026</i>
B.E. Computer Science & Eng.	Malnad College of Engineering	8.00 CGPA	<i>Jul 2024</i>
PUC (PCMC)	Brilliant P.U. College, Hassan	85%	<i>Jun 2020</i>

CERTIFICATIONS & AWARDS

- **Best Paper Award (IEEE ICANCS-2025):** For research on "Evolution of Oral Cancer Detection Techniques with VELscope and AI Integration," Bengaluru, Nov 2025.
- **Elite Gold Certifications (VTU COE):** Ethical Hacking (**97%**), Reinforcement Learning (**95%**), Social Networks (**96%**).
- **NPTEL:** Advanced coursework in Data Science for Engineers and Python-driven Data Analytics.